Charles Langer

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Education

Bachelor of Science in Electrical Engineering, GPA: 3.85 December 2026

University of Utah, Salt Lake City

- Seven-Time Dean's List
- Co-Author, IEEE Manuscript Submission

Relevant Coursework

- Classical Control Systems
- Linear & Active Circuit Design
- Cyber-Physical Systems
- Power System Analysis
- Introduction to Power
- Object-Oriented Programming
- Fundamental Electromagnetics
- PDEs for Engineers
- Power Economics/Operations

Technical Skills

- Programming & Modeling: MATLAB, Python, Java, AI Prompt Engineering
- Electrical Design: AutoCAD, Revit, One-Line Diagrams, Raceway & Cable Schedules, Control Panel Layouts
- Power & Controls: Power System Analysis, PLCs, Ladder Logic, PID Control, SCADA, Protection & Relaying
- Engineering Standards: NFPA 70 (NEC), IEEE Standards, NFPA 130
- Microsoft Office**:** Word, PowerPoint, Excel

Work Experience

BAMBC Controls May 2026 – August 2026

Electrical Engineering Intern Burbank, CA

- Translated mechanical requirements into PLC-based control logic for a dual sump pump system
- Engineered and built a low-cost control system prototype, including panel layout, wiring, PLC input and output connections, switches, and status indicators
- Formulated lead/lag sequencing and interlocks using simulated field inputs to detect pump-flow failures
- Devised and executed a testing procedure to verify PLC inputs, outputs, wiring, simulated field signals, interlocks, and ladder-logic operation

Technical Projects

Hospital Electrical Distribution Design Fall 2026 | In Progress

- Developing a hospital electrical one-line diagram incorporating utility service, transformers, emergency generators, ATS, and UPS systems
- Calculating preliminary electrical loads and modeling electrical distribution equipment and pathways in Revit
- Applying relevant NEC requirements and NFPA 99 and NFPA 110 standards to guide system design

Traction Power Substation Design Spring 2026

- Designed and reviewed single-line diagrams for AC utility interconnection and 750 V DC traction power systems, including Delta-Wye transformers and rectifier conversion
- Assisted in interpreting electrical drawings and IEEE protective relay device numbers for breakers, fuses, instrument transformers, and multifunctional protective relays to ensure proper system coordination and code compliance
- Developed TPSS equipment layout and raceway/cable schedule for utility, transformer, rectifier, and feeder connections
- Sketched a Blue Light Station with an Emergency Trip System integrating PLCs to provide remote shutdown capability and coordinate protective devices with SCADA communications

Modeling & Controlling of a Flexible System Fall 2025

- Worked with a 5-person team to model a flexible crane system in MATLAB using Bode plots and dynamic analysis
- Programmed and tuned a PID controller using root locus, phase margin, and steady-state error analysis to meet settling time and overshoot specifications
- Facilitated project coordination and consolidated modeling and control results into a formal technical report

Affiliations

- Institute of Transportation Engineers Student Chapter Member July 2025 – Present
- Vice-President of University of Utah Badminton Club May 2024 – Present